Learning for Data Science

Course 14: Unsupervised Learning

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October 13, 2026



Outline

How to Learn with no supervision

PCA: Principal Component analysis

Clustering

Hierarchical Clustering

How to Learn with no supervision

What is unsupervised learning?

- Supervised learning: Learn some mapping $f : X \to Y$
- Unsupervised learning:
  - *What can be done with* (x_i)
  - *Exploring the data*
- Not really a notion of true answer (unlike in the supervised learning case)

Common Themes

- Much harder to evaluate
- Most methods can be categorized as either,
  - Dimensionality reduction: Few features that summarize many
  - Clustering: Few “prototypes” that are representative of whole dataset
  - $\Rightarrow$ Find better (i.e. smaller) representations
- Canonical examples covered in this lecture,
  - PCA (dimensionality reduction)
  - K-means and Hierarchical Clustering (clustering)





How is that done?

Training Data		Machine learning algorithm		Predictions[1]
x_1		Hypothesis function		New example x
x_2	$\longrightarrow$	$? \approx h(x_i)$	$\longrightarrow$	$? = h(x)$
x_3				
$\vdots$				

[1] Credit: Zico Kolter

How is that done?

Training Data		Machine learning algorithm		Predictions[2]
(x_1, y_1) (x_2, y_2) (x_3, y_3) $\vdots$	$\longrightarrow$	Hypothesis function $y_i \approx h(x_i)$	$\longrightarrow$	New example x $\hat{y} = h(x)$

[2] Credit: Zico Kolter

How is that done?

- In practice, similar ML ideas are used:
  1. Define the hypothesis function (what is being extracted from the data)
  2. Define a loss function
  3. Minimize the loss

Unsupervised learning framework

- **Input features:** $x_i \in \mathbb{R}^n,\ i = 1, \ldots, m$
- **Model parameters:** $\theta \in \mathbb{R}^k$
- **Hypothesis function:** $h_\theta : \mathbb{R}^n \to \mathbb{R}^n$, approximates input given input, i.e. the goal is $x_i \approx h_\theta(x_i)$
- **Loss function:** $\ell : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}_+$, measures the difference between a hypothesis and actual input, e.g.: $\ell(h_\theta(x), x) = \|h_\theta(x) - x\|_2^2$
- Similar canonical machine learning optimization as before:

$$\min_\theta \sum_{i=1}^{m} \ell(h_\theta(x_i), x_i)$$

Unsupervised learning framework

- What is the goal here????
- Why not simply use: $h_\theta(x) = x$?
- Identity mapping **is useless.**
- The point is to do some **compression! (smaller representations).**
  - Find clusters: (describe the data with a smaller number of groups and their representative)
  - Dimensionality reduction (find a smaller representation of x)

Dimensionality reduction: a smaller representation of the input

What is this?



Dimensionality reduction: a smaller representation of the input

What is this?

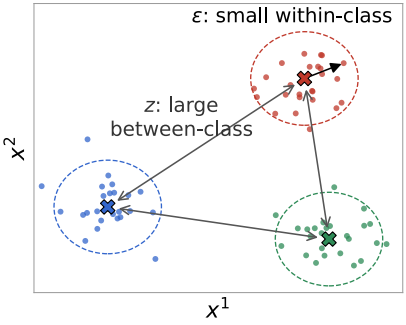


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[3] Credit for the idea: Prof. Julie Josse

A smaller representation of the input

- Idea: Certain views of high-dimensional data are **more informative than others**.
- Is there a low-dimensional representation with as much variation as possible?
- Why lots of variation? $x = z + \epsilon$
- Example (with labels):
  - z: high level type of the variable: large variation between classes
  - ϵ: variation within the class: smaller



PCA: Principal Component analysis

- Is there a low-dimensional representation with as much variation as possible?
- To implement the idea,
  - What will be the candidate family of low-dimensional representations?
    - → **Linear combinations of the features**
  - How to choose one of the many candidates?
    - → **Best reconstruction**

Principal Component analysis

- Consider the representation of x_i : (the datapoints are projected on a direction)

$$z_i = \sum_{j=1}^{d} \varphi_1^j x_i^j = \langle \varphi_1, x_i \rangle$$

- z_i provides a "summary" of x_i by linearly combining its components.
- Principal component analysis (Pearson, 1901)

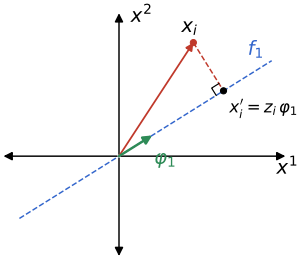
Selection Criteria: Maximal Variance

- Is there a low-dimensional representation with **as much variation** as possible?
- Optimization Problem:

$$\max_{\|\varphi_1\|=1} \frac{1}{n} \sum_{i=1}^{n} z_i^2$$

- All features should have **zero mean** (over the datapoints).
- Very nice example [here]

Selection Criteria: Maximal Variance

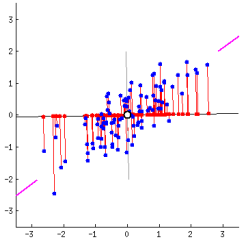


- Let f_1 be the direction of maximum variance and φ_1 a unit vector along it, i.e. $\|\varphi_1\| = 1$.
- Let x_i be any point and x_i' its projection on φ_1. Its coordinate along φ_1 is z_i:

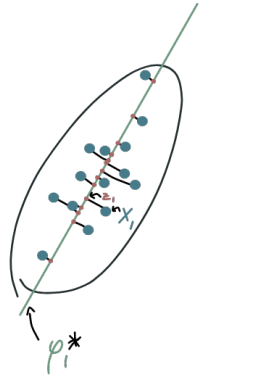
$$z_i = \frac{\varphi_1 \cdot x_i}{\|\varphi_1\|} = \varphi_1^{\mathrm{T}} x_i, \qquad x_i' = z_i \, \varphi_1$$

- The goal is to find φ_1 with $\|\varphi_1\| = 1$ maximising $\mathrm{Var}_i(\varphi_1^{\mathrm{T}} x_i) = \frac{1}{n} \sum_{i=1}^{n} z_i^2$ (zero-mean features).
- Reminder: A dot product is the overlap/similarity of the vectors.
  - The goal is a vector that overlaps well with the data (via the dot product)

Selection Criteria: Maximal Variance

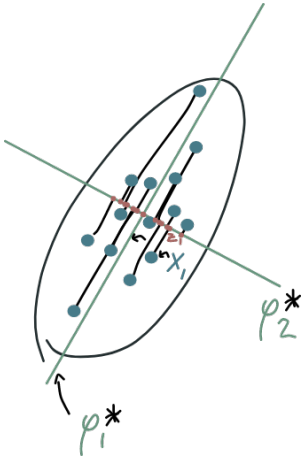






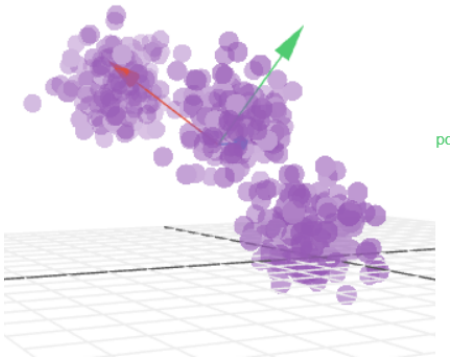
Second, third, PCA directions

- Once the first direction is found, the second main direction can be found.
- **Same problem** with the constraint of being **orthogonal to the previous direction**.
- Interpretation: PCA is finding a new, better coordinate system for the data



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- **Same problem** with the constraint of being **orthogonal to the previous direction**.
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Semantics

- φ_k: PCA "direction" or "components".
- z^k: Scores: coordinates with respect to the kth PCA direction.

Importance of Feature Normalization

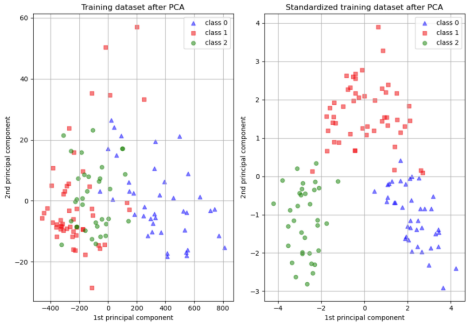
- Super important to "standardize" the features!!!
- Example:
  - Human height (1.50m ± 1m) varies less than human weight (btw 70kg ± 50kg) (just **because of the unit system!!!!!)**
  - A change of 1m is more significant than 10kgs.
- Consequence: PCA looks for correlations between the variables.
- Let X the random variable of the data such that: $x_1, \ldots, x_n \sim X$

$$\arg\max_{\|\varphi\|=1} \mathbb{E}\Big[\Big(\sum_{j=1}^{d} X^j \varphi^j\Big)^2\Big] = \arg\max_{\|\varphi\|=1} \sum_{1\leq j<l\leq d} \underbrace{\mathbb{E}[X^j X^l]\varphi^j \varphi^l}_{\text{Picks the most correlated dimensions of } X}$$

$$\mathbb{E}[X] = 0,\ \mathbb{V}[X^j] = 1 \quad \forall j$$

Importance of Feature Scaling

- Wine dataset
- Continuous features that are **heterogeneous in scale** due to differing properties that they measure (i.e alcohol content, and malic acid).
- PCA significantly more efficient **after standardization**.
- (ON THIS SPECIFIC dataset)



Link

Importance of Feature Scaling

- Standardizing the features forces the PCA to look at correlations.
  - Sometimes the features are all in the same unit
  - Dividing by std may remove the fact that some features have more "objective" variance than others.

Alternative Interpretation: Linear Approximation

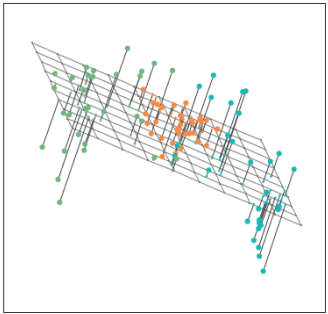
- The first K directions in PCA find the best K-dimensional linear approximation

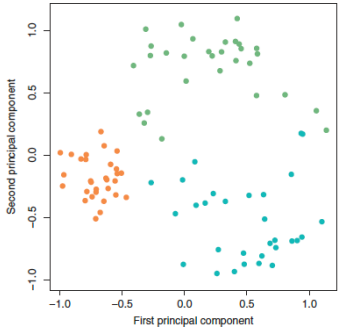
$$\min_{\varphi} \sum_{i=1}^{n} \| \underbrace{h_{\varphi}(x_i)}_{\text{of dimension } K} - x_i \|_2^2$$

$$h_{\varphi}(x) = \sum_{k=1}^{K} z^k(x) \varphi_k, \quad z^k(x) = \langle \varphi_k, x \rangle$$

Alternative Interpretation: Linear Approximation

- $h_\varphi(x) = \sum_{k=1}^{K} z^k \varphi_k$: Linear approximation of x, $\quad h_\varphi(x) = \Phi x$





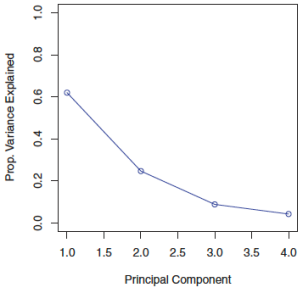
Explained Variation

Amount of variance explained by the first K principal components.

$$\frac{\sum_{k=1}^{K}\sum_{i=1}^{n}(z_i^k)^2}{\sum_{i=1}^{n}\|x_i-\bar{x}\|^2} \qquad \text{(btw 0 and 1)}$$

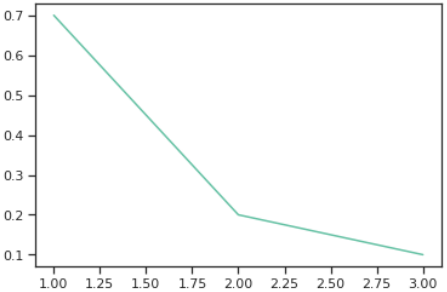
Explained Variation

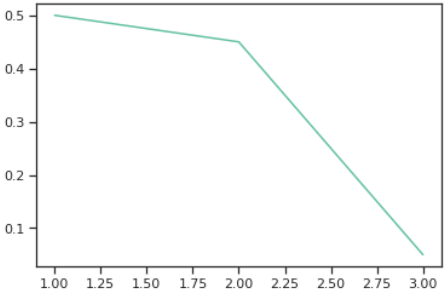
- 62% of the data variance explained by the first component
- 24.7% explained by the second component
- The first two components give a pretty accurate picture of the data variance.



Exercise

- What would the proportion of variance explained look like if:
  - The (3D) data were shaped like
    - A pancake (two long directions, one short one)?
    - A cigar (one long direction, two short ones)?





How to choose K the number of components

- No clear consensus
- Application dependent
- Good to look at the explained variations
- Rule of thumb:
  - If there is nothing interesting in the first component then it is most likely useless to look further
  - If there is something interesting then increase K
    until there is nothing more interesting

Handling missing Values

- Reconstruction problem with missing entries:

$$\min_{\varphi} \sum_{i=1}^{n} \sum_{j=1}^{d} \left([h_\varphi(x_i)]^j - x_i^j \right)^2$$

$$\Downarrow$$

$$\min_{\varphi} \sum_{(i,j)\in\mathcal{O}} \left([h_\varphi(x_i)]^j - x_i^j \right)^2, \qquad \mathcal{O} : \text{observed entries}$$

- Once solved it can be used to predict the missing entries !!!

Application: Netflix challenge

- Dataset:
  - $n = 480{,}189$
  - $p = 17{,}770$
  - 99% of missing values.
- Goal: suggest relevant movies.
- 1M USD challenge
- Idea: few features should predict the taste of a customer.
- Find these features via PCA

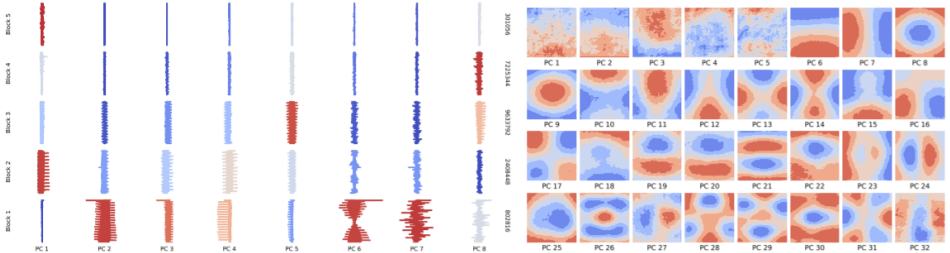
	Jerry Maguire	Oceans	Road to Perdition	A Fortunate Man	Catch Me If You Can	Driving Miss Daisy	The Two Popes	The Laundromat	Code 8	The Social Network	⋯
Customer 1	•	•	•	•	4	•	•	•	•	•	⋯
Customer 2	•	•	3	•	•	•	3	•	•	3	⋯
Customer 3	•	2	•	4	•	•	•	•	2	•	⋯
Customer 4	3	•	•	•	•	•	•	•	•	•	⋯
Customer 5	5	1	•	•	4	•	•	•	•	•	⋯
Customer 6	•	•	•	•	•	2	4	•	•	•	⋯
Customer 7	•	•	5	•	•	•	•	3	•	•	⋯
Customer 8	•	•	•	•	•	•	•	•	•	•	⋯
Customer 9	3	•	•	•	5	•	•	1	•	•	⋯
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

Conclusion: Practical tips

- Even though the method is easy to run, there are lots of potential issues,
  - Variables might be at different units, and there might be ambiguity about whether to rescale them.
  - The directions are only **unique up to sign** (be careful when interpreting the sign of the values z_i)
  - Can handle missing values!
  - Choosing K may be tricky.

Still recent advance on PCA

- New techniques to compute the PCA of the activation of a deep net (ResNet 200) on Imagenet.
- $d = 20\text{M}$
- $n = 1.2\text{M}$
- The covariance matrix is 195TB (of course will never fit in memory....)



(a) Principal Components (b) Block-1 Mean Filter Maps

Link

Recent Methods for Large Scale UL

- CLIP: Connecting text and images (Radford *et al.*, 2021)
- What if there is some shared structure across partitions of x?
- $X = (X_{\text{image}}, X_{\text{text}})$
  - They both describe the same concept.
- Then the goal is
  - $f_{\text{image}}(X_{\text{image}}) \approx f_{\text{text}}(X_{\text{text}})$

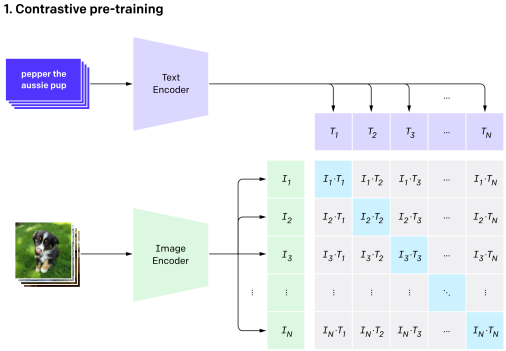


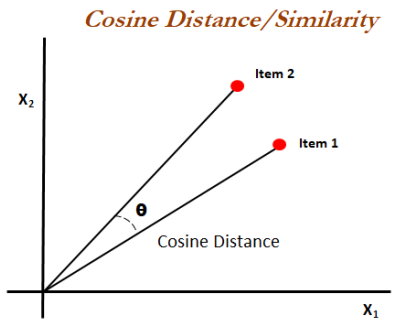
4

[4] https://openai.com/index/clip/

CLIP: Connecting Text and Images

- Then the goal is
  - $f_{\text{image}}(X_{\text{image}}) \approx f_{\text{text}}(X_{\text{text}})$





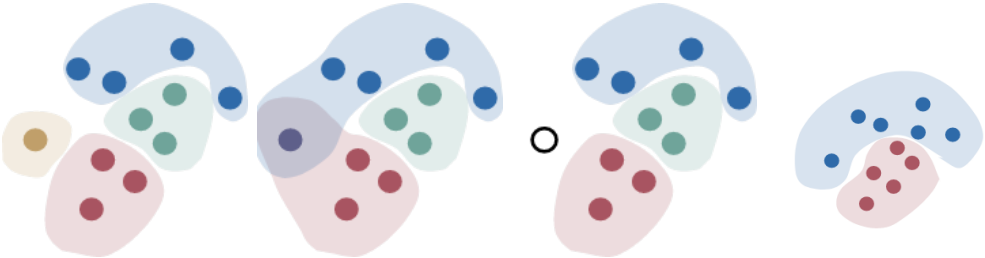
Cosine Similarity

Clustering

- Idea: Partition the observations, so that those that are similar to each other appear together
- Look for homogeneous subgroups in heterogeneous data.

Formalization

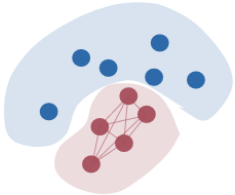
- A partition $C_1, \ldots, C_K$ is a collection of subsets satisfying,
  - Each sample is in a subset: $C_1 \cup \cdots \cup C_K = \{1, \ldots, n\}$
  - Subsets are disjoint: For any pair, $C_k \cap C_{k'} = \emptyset$ for $k \neq k'$



Formalization of Clustering

The goal is not just any partition, but the one that **minimizes within group variation**, called $W(C_k)$:

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} W(C_k)$$



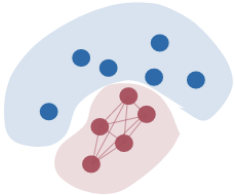
Usually:

$$W(C_k) = \frac{1}{|C_k|} \sum_{i,i' \in C_k} \underbrace{\|x_i - x_{i'}\|^2}_{\text{Sum of distances}}$$

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Minimize over subsets: Intractable

- An algorithm is needed to find an approximate solution

K-Means

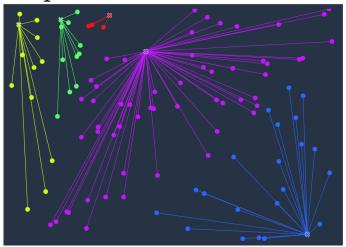
- This is a combinatorial optimization problem, and finding the global optimum is computationally challenging.
- However K-means (Lloyd, 1982) usually finds good local optima,

1. Arbitrarily assign each x_i to one of the clusters, $C_1, \ldots, C_K$.
2. Iterate until convergence,
   a. Compute the mean $\bar{x}_k$ of the points in C_k.
   b. Reassign points x_i so they are put in the cluster whose centroid they are closest to.

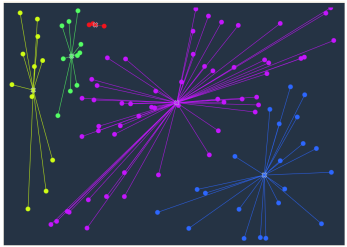
K-means

Very nice Demo: [here]

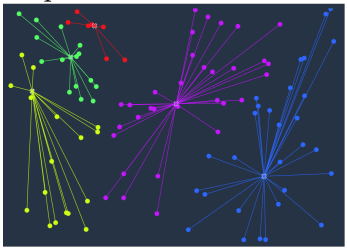
Step 1:



Step 2:

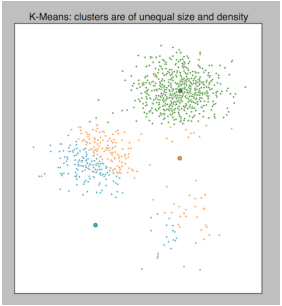


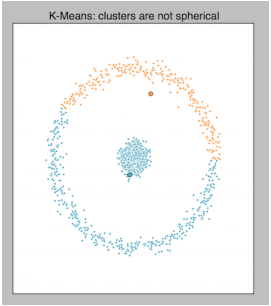
Step 3:



Limitations

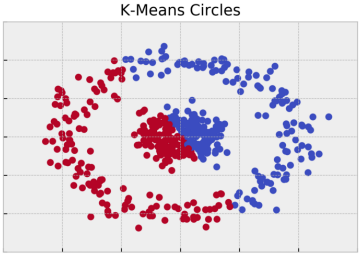
- K-means often does not do a good job in clustering when there are:
  - Variations in density (clusters of different sizes)
  - **Nonspherical shapes of clusters (because of the distance metric used)**
  - Outliers.

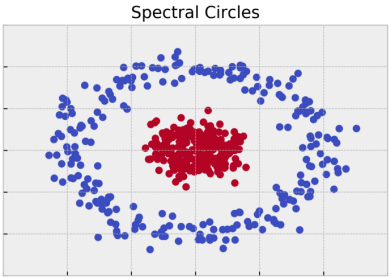




How to address (some of) the limitations

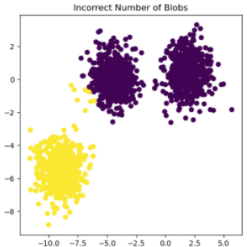
- Try different notions of distances (see following)
- Extract better features to do the clustering. (e.g. polar coordinattes or main component of the Laplacian matrix) [see here for more details]





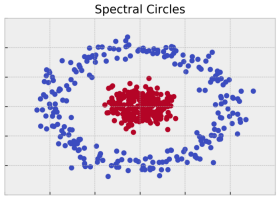
Supervised Evaluation of K-means

- **Supervised evaluation:** use a pre-classified dataset as a benchmark.
- Not completely trivial:
  - Value of the **labels are arbitrary**.
  - Maybe not the right number of clusters.
  - Can **not** do 0-1 loss on the classes.



Link

Red = 1, Blue = 0



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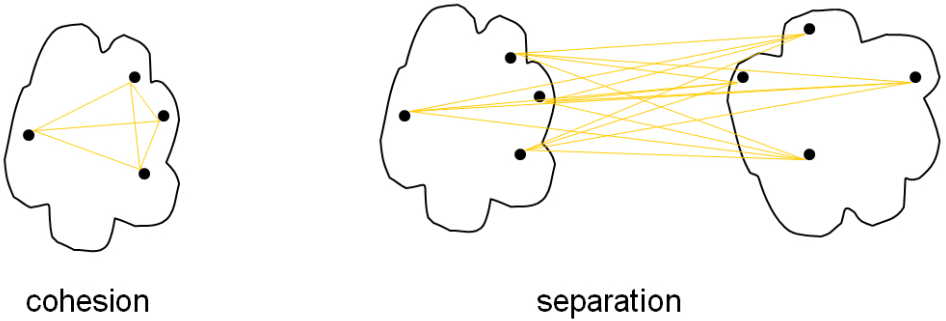
Supervised Evaluation of K-means

- **Supervised evaluation:** use a pre-classified dataset as a benchmark.
- Not completely trivial: Value of the **labels are arbitrary** + Maybe not the right number of clusters. Can **not** do 0-1 loss on the classes.
- **Homogeneity:** each cluster contains only members of a single class.
- **Completeness:** all members of a given class are assigned to the same cluster
- Many measures available on sklearn (with their pros and cons)

```
>>> from sklearn import metrics
>>> labels_true = [0, 0, 0, 1, 1, 1]
>>> labels_pred = [0, 0, 1, 1, 2, 2]
>>> metrics.rand_score(labels_true, labels_pred)
0.66...
>>> labels_pred = [1, 1, 0, 0, 3, 3]
>>> metrics.rand_score(labels_true, labels_pred)
0.66...
```

Unsupervised Evaluation

- Similar idea but with distance instead of labels
  - Minimize intra-cluster distance (**maximize intra-cluster cohesiveness**)
  - Maximize inter-cluster distance (**maximize inter-cluster separation**)



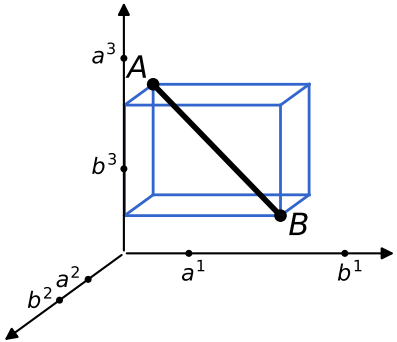
Euclidean distance

In general in $\mathbb{R}^d$:

$$D(A,B) = \sqrt{\sum_{j=1}^{d} (a^j - b^j)^2}$$

Usual notation:

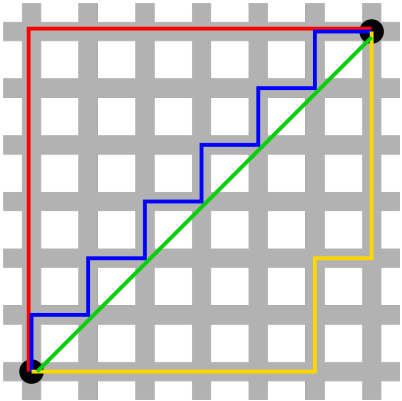
$$D(A,B) = \|a - b\|_2$$



Manhattan Distance

- **Euclidean Distance: 8.5**
- **Manhattan Distance: 12.0**
- **Manhattan Distance: 12.0**
- **Manhattan Distance: 12.0**

$$d_1(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_1 = \sum_{j=1}^{n} |p^j - q^j|$$



Hamming Distance

- For sequences (symbols, bits) with the **same length.**
- Hamming distance: **number of characters to modify** to go from X to Y:

$$H(x, y) = |\{j : x^j \neq y^j\}|$$

- Example:

```
x = intelligence
y = ixteuligncee
```

$$H(x, y) = 5$$

Hamming Distance

a	b	$d_H(a, b)$	`len(a)`
`"0000"`	`"1111"`	4	4
`"1101"`	`"1111"`	1	4
`"2173896"`	`"2233796"`	3	7
`"hydroxychloroquine"`	`"idroxichloroquinne"`	16	18

Levenshtein Distance

- For sequences (symbols, bits) with **a different length.**
- **Number of characters to modify add or remove** to go from X to Y:
  - Replace: $x^j \rightarrow a$
  - Add: $x^j \rightarrow x^j \cdot a$
  - Remove: $x^{j-1} \cdot x^j \cdot x^{j+1} \rightarrow x^{j-1} \cdot x^{j+1}$
- Example:

```
x = intelligence
y = ixteulignceee
```

$$E(x, y) = 5$$

Levenshtein Distance

a	b	$d_L(a,b)$	$d_H(a,b)$	`len(a)`
`"0000"`	`"1111"`	4	4	4
`"1101"`	`"1111"`	1	1	4
`"2173896"`	`"2233796"`	3	3	7
`"hydroxychloroquine"`	`"idroxichloroquinne"`	4	16	18

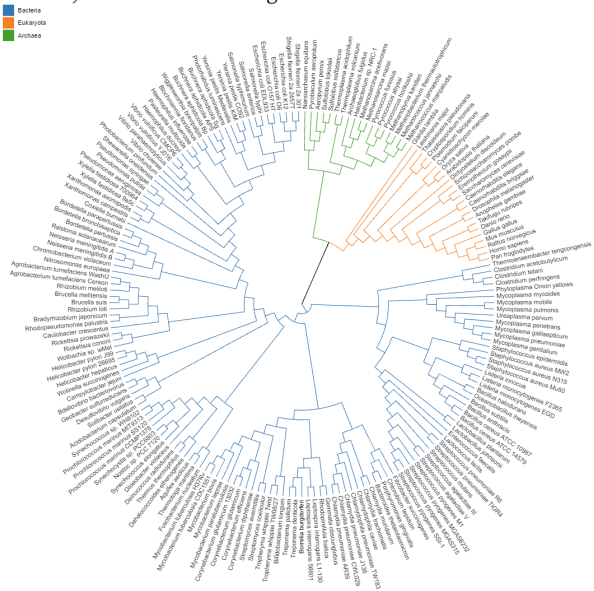
Hierarchical Clustering

- K controls the "magnification" at which the clustering is done.
- What if the clustering could be done at many different scales, all at once?
- Hierarchical of clusters defined by using a tree.

Hierarchical clustering

Link

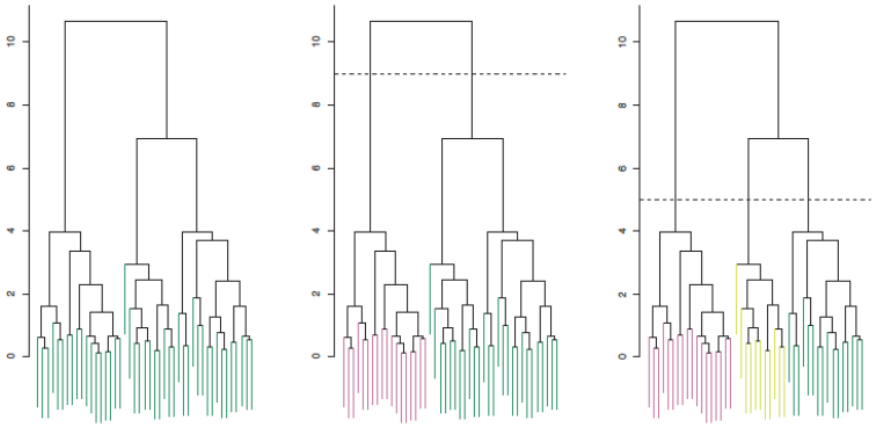
Behold, the cluster dendrogram.



Interpretation

- Samples which are similar to each other are put on the same subtree.
- Pairs of samples that are very similar to one another
  share very recent common ancestors.
- Beware: Samples can be close by at the leaves without being close in the subtree sense.
- You can get a standard clustering by "cutting" tree at some horizontal level

Interpretation: cutting the tree



That's cool, how are these trees made?

- These trees are informative. An automated procedure for creating them is needed.
  - **Agglomerative:** bottom-up
  - **Divisive:** top-down
- Agglomerative:
  1. **Initialize:** Associate each point with a cluster
  2. **Iterate until only one cluster:** Look at all pairs of clusters. Merge the pair which are the most similar.



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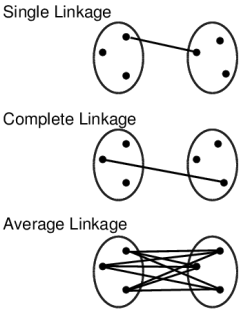
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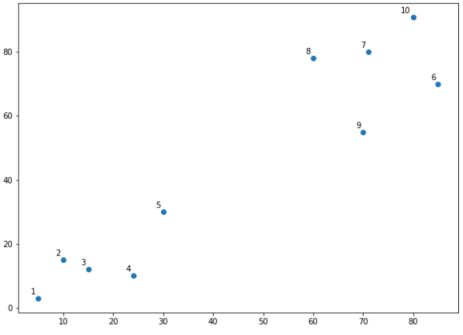
Similarity between clusters

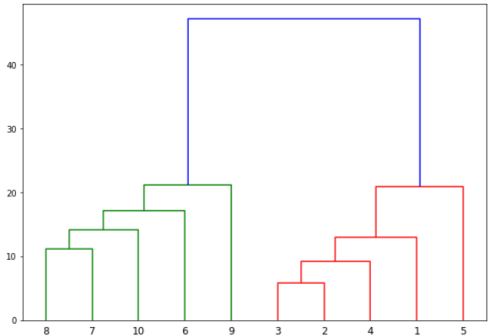
- The height on the dendrogram gives the similarity between descendants.
- But what's a good distance between pairs of sets?
  - **Single:** Minimum distance between any pair of points
  - **Complete:** Maximum distance between any pair of points
  - **Average:** Average distance over all pairs of points



Similarity between clusters

- The distance between the clusters at any iteration can be visualized on the tree
- Merges lower on the tree –> Smaller intercluster distance





Link

Top-Down Clustering

- Divisive:
  1. **Initialize:** start with a cluster with all the points.
  2. **Iterate until only one point per cluster:** For each cluster: With any clustering method (e.g. K-means) create K sub-clusters.

Comparison between K-means and hierarchical clustering

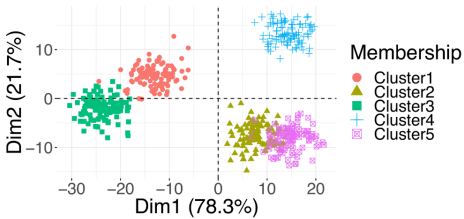
1. K-means can handle big data more efficiently compared to hierarchical method.
2. K-means clustering requires prior knowledge of K. But, you can stop at whatever number of clusters you find appropriate in hierarchical clustering by interpreting the dendrogram.
3. K-Means is found to work well when the shape of the clusters is hyper spherical whereas hierarchical clustering works better in case of non-spherical data distribution. (hierarchical clustering is more local)
4. Results are deterministic in Hierarchical clustering, whereas they might differ in K-means due to random choice of initial clusters

Practical tips

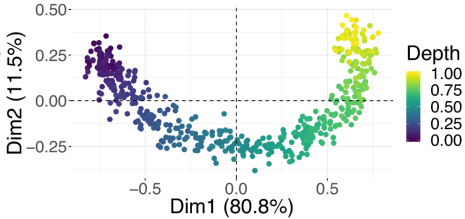
- Clustering: method for "grouping similar datapoints".
- Many algorithms to do that:
  - This lecture covered **k-means clustering** but there are many other techniques.
  - The choice of distance is crucial.
    - Numeric data: e.g., Euclidean, Squared Euclidean, Manhattan, Mahalanobis, Minkowski
    - Text or other non-numeric data: Hamming, Levenshtein
  - Don't try to find a "perfect" distance.
  - Try many, and assess the sensitivity of the findings.
  - Choosing K may be once again tricky and lead to drastically different results

Conclusion: Objective of dimensionality reduction

- Compress data while preserving most of the meaningful information
- What are the latent patterns in the data?
  - Discrete change in attributes: clustering
  - Gradual shift in pattern: gradients



(a) Clusters



(b) Gradient

Additional Reading

- ISLR book: chapter 12
- Probabilistic machine learning: section V

Refs used to build this course

- Zico Kolter's course on Data Science.
- https://medium.com/analytics-vidhya/mathematics-behind-principal-component-analysis-pca-1cdff0a808a9
- https://builtin.com/data-science/step-step-explanation-principal-component-analysis

Stuart Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.

Karl Pearson. On lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 2(11):559–572, 1901.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, pages 8748–8763, 2021.